Geometry-Only Identifiability of Lunar Prospector Spatiotemporal Surface-Potential Reconstructions

Experiment 008

13 July 2026

Abstract

We prospectively audited whether the Lunar Prospector electron-reflectometer sampling geometry can identify a global, time-dependent lunar potential field. No fitted potential, fit-quality flag, beam estimate, or spacecraft-potential estimate entered the analysis. Each structurally valid, surface-connected 15-row energy sweep was granted an oracle-known scalar spacecraft-relative potential, making the experiment an optimistic upper bound on information in any later selected measurement set. The primary model comprised independent 24-hour real spherical-harmonic fields through degree $\ell = 5$ (36 parameters per day). All 560 daily operators were algebraically full rank, but none was physically stable: the median sampled-RMS/global-RMS visibility of the weakest mode was $4.5191538242914006 \times 10^{-5}$, and all 560 windows failed the frozen 25-V recovery gate. In contrast, prospectively declared 72-hour $\ell = 1$, 168-hour $\ell \leq 3$, one-synodic-period $\ell \leq 10$, whole-mission static $\ell \leq 20$, and whole-mission constant-plus-synodic $\ell \leq 10$ classes passed their applicable stability and recovery gates. The frozen decision tree therefore yields **Outcome C**: the submitted daily $\ell = 5$ class fails, while specified coarse, static, and shared-synodic classes are conditionally identifiable from the design. This is not recovery of absolute lunar surface potential. Measurement validation remains open, and an unknown additive spacecraft potential is exactly confounded with the spatial monopole and its temporal counterparts.

1 Background and scope

Lunar Prospector sampled along a moving orbital track rather than observing the whole lunar surface simultaneously. A regularized spherical-harmonic fit can always return a smooth map, but smoothness does not establish that the data constrain that map. Experiment 008 therefore asks a narrower question: given the actual sampling times and corrected magnetic-field surface intersections, which declared spatial and spatiotemporal field classes are identifiable?

The protocol, configurations, seeds, model grid, thresholds, and outcome tree were frozen before any scientific singular spectrum or recovery result was opened. Two independent pre-outcome audits found that empirical column normalization, an unweighted latitude–longitude grid norm, midpoint temporal evaluation, Gram-matrix rank decisions, and 32-draw tail estimates could give misleading answers. Prospective Amendment 03 replaced them with a physical field metric, exact row-level temporal products, direct null-space checks, tall-skinny QR (TSQR), and a 16,384-draw covariance gate. It also made the spacecraft-potential gauge and the outcome precedence explicit. The original 32-replicate injections and empirical column-normalization diagnostics were retained as labeled checks rather than discarded.

The information barrier allowed the 560 raw Level-1 ER/MAG files, attitude history, SPICE kernels, and geometry/harmonic code. It prohibited fitted potentials and all selections derived from

them. Thus, a failure here is an upper-bound sampling result: removing sweeps after fit-quality selection cannot add geometric information. A success here is only a design result and does not validate a measurement.

2 Estimand, observation unit, and gauge

The canonical estimand grants sweep i one noisily observed scalar relative potential and represents a declared field by coefficients c:

$$y_i = A_i c + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

This is an oracle measurement model, not a claim that the loss-cone analysis has already delivered such a scalar. One sweep contains 15 energy rows, but those rows share one fitted potential. The observation operator consequently has one row per sweep,

$$A_{i,qk} = \frac{1}{n_i} \sum_{j \in i} T_q(t_{ij}) R_k(x_{ij}), \tag{2}$$

where the sum includes only surface-connected row footprints. The midpoint UTC used to assign a sweep to a window is computed from all 15 structurally valid UTC rows. Equation (2) evaluates the temporal–spatial product at every connected row; it is not $T_q(t_{i,\mathrm{mid}})\,\overline{R_k(x_i)}$.

The scalar available to a surface-connected electron measurement is spacecraft-relative. If an additive spacecraft potential $u(t)$ is unknown, then adding offset columns to Equation (1) duplicates the spatial $\ell = 0$ column and, for a temporal model, its matching temporal columns. The augmented absolute-potential problem is therefore exactly rank deficient in those directions. We report both the optimistic oracle estimand retaining $\ell = 0$ and a gauge-safe contrast estimand with every $\ell = 0$ column removed. Neither can justify the phrase "recovered absolute surface potential."

3 Geometry and the production-frame defect

3.1 Canonical extraction

A sweep was retained only if it was an exact contiguous 15-row group, every UTC was finite, and every MAG vector was finite with $10^{-9} < \|B\| < 10^{3}$. Canonical geometry uses circular right-ascension and linear declination interpolation, constructs the spacecraft-despun to J2000 rotation from a J2000 spin axis and J2000 spacecraft-to-Sun vector, rotates MAG into `IAU_MOON`, and applies the frozen ray–sphere intersection. At least one finite intersection admits a sweep to the optimistic population. Table 1 gives the exact mission census.

3.2 Frame mismatch discovered before outcomes

Static code inspection found a production coordinate-frame defect distinct from the earlier next-sample attitude lookup. The production path rotated the spacecraft-to-Sun vector from J2000 to `IAU_MOON`, then passed it with a J2000 attitude-derived spin axis to a rotation builder whose two inputs are defined to be J2000 vectors. Projecting one vector against another expressed in a different frame is not a valid coordinate operation. Amendment 01 froze four variants before operator outcomes: canonical frame-consistent geometry; frame-consistent geometry with legacy next-sample attitude; corrected attitude with the frame-mixed production path; and the submitted combination of legacy attitude plus frame mixing.

Table 1: Geometry-only input census. All 560 files processed without error.

Quantity	Count
Raw rows	8,389,815
Contiguous groups / exact 15-row sweeps	559,321
Malformed sweep groups	0
Structurally valid sweeps	551,637
Canonical connected sweeps	399,643
Canonical complete-geometry sweeps	376,920
Legacy-attitude connected / complete	399,647 / 376,962
Frame-mixed connected / complete	421,502 / 399,680
Submitted-legacy connected / complete	421,545 / 399,663

On the pre-outcome one-day fixture, the frame-mixed implementation reproduced production with 9,180 valid projected rows, zero validity or polarity disagreements, and maximum latitude and longitude discrepancies of $2.2737367544323206 \times 10^{-13}$ and $5.115907697472721 \times 10^{-13}$ degrees. Relative to canonical geometry, the production frame mix moved common footprints by 47.8247 km median, 194.768 km at the 95th percentile, and 419.166 km maximum. The attitude-only contrast was 1.06186, 9.75790, and 85.2621 km; the submitted combined legacy contrast was 47.9543, 194.283, and 417.739 km. The defect is therefore material to production footprints and connectivity, although, as shown below, it does not change the primary identifiability verdict.

4 Real harmonics and the physical field metric

Columns are degree-major and order-minor, $m = -\ell, \ldots, +\ell$. From SciPy's orthonormal complex harmonics $Y_{\ell m}$ with the Condon–Shortley phase, the real basis is

$$R_{\ell m} = \begin{cases} \sqrt{2}(-1)^{|m|} \operatorname{Im} Y_{\ell,|m|}, & m < 0, \\ Y_{\ell 0}, & m = 0, \\ \sqrt{2}(-1)^{m} \operatorname{Re} Y_{\ell m}, & m > 0. \end{cases} \tag{3}$$

The implementation was tested for numerical orthonormality and for field and coefficient-norm parity with the repository's complex representation. A degree limit L has $p = (L+1)^2$ spatial coefficients.

For $f(x) = \sum_k c_k R_k(x)$, the analytic surface-area mean-square metric is

$$\|f\|^2_{\text{global}} = \frac{1}{4\pi} \int_{S^2} f(x)^2 \, d\Omega = c^T G_x c, \qquad G_x = \frac{I}{4\pi}. \tag{4}$$

This supersedes the original unweighted equal-angle grid, which overweights polar samples. An area-weighted 2-degree grid was used as a discretization check; the unweighted grid remains only a legacy replay.

The shared synodic model uses $T(t) = (1, \cos \omega t, \sin \omega t)$ with period 29.530588 days. Its metric is $G = G_t \otimes G_x$, where G_t is the analytic continuous-time average of $T(t)T(t)^T$ over the half-open interval from the earliest complete-sweep UTC through one nanosecond beyond the latest. Static and independent-window models use G_x.

Let $d = G^{1/2}c$ be metric coordinates, so $\|d\|_2$ is global RMS. The primary decomposition is

$$\widetilde{A} = \frac{AG^{-1/2}}{\sqrt{n}} = U\Sigma V^T. \tag{5}$$

Each singular value is consequently a sampled-RMS/global-RMS visibility ratio. A right singular vector maps to physical coefficients through $G^{-1/2}$. This makes a weak column or weak combination physically visible in the spectrum instead of scaling it away by empirical unit-column normalization.

5 Numerical audit and decision rules

Independent-window operators were decomposed by direct float64 SVD. Large whole-mission operators were streamed in chunks: each chunk received a reduced QR factorization; stacked R factors were repeatedly reduced; and the final small $R/\sqrt{n}$ received a direct SVD. This TSQR construction preserves the tall matrix's singular spectrum without squaring its condition number. Gram matrices were not used for rank or effective-rank decisions.

Exact numerical rank used $\max(n, p)\epsilon_{64}\sigma_{\max}$. Effective ranks used relative thresholds 10^{-3}, 10^{-6}, and 10^{-9}. A window was unstable if any of the following held: exact rank below p; condition number at least 10^6; effective rank at 10^{-6} below p; or minimum physical visibility at most 0.1. A 100-V global-RMS weakest mode is a material near-null witness when at least 50% of eligible windows have visibility at most 0.1. Nonempty epoch- anchored half-open windows were eligible; empty scheduled windows were reported separately, and a nonempty underdetermined window was an attempted failure.

The primary class passes only if at least 90% of windows are stable, at least 90% pass recovery, and no material witness exists. Outcome B requires at least 50% primary instability, at least 50% recovery failure, and a material witness. Outcome C takes precedence over B when all B criteria hold and at least one prospectively declared coarse or static class passes the analogous 90% gates without a material witness. Provenance, geometry, or numerical stop-rule failure would instead force Outcome D.

6 Recovery, challenge injections, and blocked tests

6.1 Frozen recovery gate

For a full-rank unregularized estimator with independent sweep noise, the metric-coordinate estimation error is independent of injected truth. If s_j are the singular values in Equation (5), its covariance eigenvalues are $\sigma^2/(ns_j^2)$. The outcome gate evaluated this exact covariance with 16,384 deterministic PCG64DXSM standard-normal draws per parameter count at $\sigma = 25$ V. Relative global error was divided by a 100-V truth RMS. A window passed only when its median error was below 0.25 and its 90th percentile was below 0.5; rank-deficient windows failed automatically.

The original explicit trials remained an implementation check: 32 PCG64 realizations in every window at 10, 25, 50, and 100 V, with degree power proportional to $(\ell+1)^{-2}$ and every truth scaled to 100 V global RMS. Equal-total-power degree shells and the weakest metric mode challenged spectral dependence. Recoveries included unregularized SVD, relative truncated SVD at 10^{-3}, 10^{-6}, and 10^{-9}, and ridge penalties 10^{-6}, 10^{-3}, 1, and 10^3 after empirical column normalization. Regularized answers are bias–variance diagnostics and cannot establish unregularized identifiability.

Six out-of-basis great-circle Gaussian caps used centers $(0, 0)$, $(60, 0)$, and $(-60, 180)$ degrees and angular FWHM 15 and 30 degrees. Their errors were decomposed into best-$L = 5$ approximation

bias and estimation error. These cap tests were prospectively descriptive and do not enter Outcomes A–D.

6.2 Blocked synthetic recovery

Blocked tests generated observations before splitting and kept complete sweeps together. Whole-mission static and constant-plus-synodic $L = 5$ models used a final-25%-of-time holdout, twelve 30-degree longitude holdouts, and four interleaved seven-day-block folds. Daily independent fields instead used a contiguous final-quarter holdout within each day and every available 30-degree longitude sector. All blocked tests are descriptive: the mission- tail test is not meaningful for unrelated daily coefficient vectors, and success on matched synthetic truths does not show that the real field belongs to the shared temporal class.

7 Results

No provenance, geometry, replay, or numerical stop rule was triggered. The canonical geometry cache contained 399,643 connected sweeps, and the primary calendar-day partition had 560 nonempty windows and zero scheduled empty windows.

7.1 Primary daily degree-5 model

Table 2 is the central result. Every daily matrix has numerical rank 36/36. Thus there is no exact algebraic null space under the rank tolerance. Nevertheless, all 560 matrices have a physically weak mode far below the 0.1 visibility threshold. Algebraic full rank is not physical identifiability at the declared field amplitude and noise scale.

Table 2: Primary independent 24-hour, $L = 5$ audit. Recovery values are relative global-RMS errors at 25-V sweep noise.

Diagnostic	Result
Eligible / empty windows	560 / 0
Rows per window	205–967
Exact rank 36/36	560/560
Minimum visibility, min / median / max	$1.085048307712986 \times 10^{-6}$ / $4.5191538242914006 \times 10^{-5}$ / $4.0496025853511009 \times 10^{-4}$
Condition number, min / median / max	5,008.16191 / 47,125.23367 / 1,934,950.62826
Condition $\geq 10^6$ / effective-rank-10^{-6} failure	11 / 11
Effective-rank-10^{-3} failure	560
Stable / material-witness windows	0 / 560
Covariance median-error range	16.9976255298–6,373.24934397
Covariance 90th-percentile range	38.8118610674–15,401.4657774
Recovery-passing windows	0/560

Scaling the weakest metric mode in each window to exactly 100 V analytic surface-area RMS gave the following minimum, median, and maximum sampled RMS:

$$\left(1.08504830771 \times 10^{-4},\ 4.51915382429 \times 10^{-3},\ 4.04960258535 \times 10^{-2}\right)\ \text{V}.$$

Direct replay agreed with $100\sigma_{\min}$ to numerical precision. The old unweighted equal-angle grid assigned these same witnesses only about 81 V RMS, demonstrating why it was not retained as the physical metric. The daily operators are therefore algebraically full rank but not physically identifiable under the frozen visibility and recovery definitions.

7.2 Temporal-resolution boundary

Failure was not universal across all parameterizations. No tested class at one or six hours passed. At 24 hours, $L = 1$ recovered noisy injections in all 560 windows but only 435/560 (77.6786%) were stable, below the 90% class gate. The first passing class was the 72-hour $L = 1$ model. Table 3 reports the decision boundary and all passing independent-window classes.

Table 3: Independent-window boundary. Fractions are stable, recovery-passing, and weakest-visibility ≤ 0.1 witness fractions.

Hours	L	Windows	Stable	Recovery	Witness	Class
24	1	560	0.776786	1.000000	0.223214	fail
24	5	560	0.000000	0.000000	1.000000	fail
72	1	187	1.000000	1.000000	0.000000	pass
168	1	81	1.000000	1.000000	0.000000	pass
168	2	81	0.987654	0.987654	0.012346	pass
168	3	81	0.975309	0.987654	0.024691	pass
708.734112	1	20	1.000000	1.000000	0.000000	pass
708.734112	2	20	1.000000	1.000000	0.000000	pass
708.734112	3	20	0.950000	1.000000	0.050000	pass
708.734112	5	20	0.950000	0.950000	0.050000	pass
708.734112	10	20	0.950000	0.950000	0.050000	pass

The 708.734112-hour duration is one 29.530588-day synodic period. For $L = 5$ and 10, 19 of 20 windows passed and the single short edge window failed. These are independent piecewise-constant fields over long windows, not a continuous global history.

7.3 Static and shared-synodic controls

All prospectively specified whole-mission static controls passed through $L = 20$ (Table 4). Their strong visibility shows that the mission contains abundant integrated spatial coverage; this does not repair the lack of simultaneous daily coverage.

Table 4: Whole-mission static controls; all are full rank, stable, and pass 25-V covariance recovery.

L	Parameters	Condition	Min. visibility	Median rel. error	P90 rel. error
1	4	1.38615	0.871539	0.000744337	0.00114765
2	9	1.65853	0.837292	0.00120426	0.00161642
3	16	1.88882	0.818785	0.00164752	0.00205996
5	36	2.26619	0.799284	0.00254016	0.00295569
10	121	2.96840	0.769662	0.00475333	0.00516575
20	441	3.88098	0.726022	0.00932556	0.00975176

The mission-shared constant-plus-synodic class also passed at every specified degree (Table 5). At $L = 5$, removing the three monopole temporal columns left 105 contrast parameters and changed the condition, minimum visibility, median error, and 90th-percentile error only to 2.40114, 0.766700, 0.00436844, and 0.00477675; that contrast model also passed. All constant and constant-plus-synodic contrast controls through $L = 10$ passed.

Table 5: Whole-mission shared constant-plus-synodic oracle controls. Each has three temporal coefficients per spatial harmonic and passes all gates.

L	Parameters	Condition	Min. visibility	Median rel. error	P90 rel. error
1	12	1.43712	0.853315	0.00137446	0.00177743
2	27	1.72202	0.818734	0.00214701	0.00255761
3	48	1.97788	0.794083	0.00290868	0.00331821
5	108	2.40469	0.765616	0.00443060	0.00484974
10	363	3.20476	0.726145	0.00826680	0.00868253

The shared-synodic result is conditional on one coefficient set multiplying a constant, a synodic cosine, and a synodic sine over the entire mission. It is not evidence for freely varying daily coefficients and is not evidence that the real potential follows this temporal law.

7.4 Explicit injections and regularization

The explicit 32-replicate broadband injections agreed with the covariance gate in the outcome-determinative sense: all 560 daily windows failed both. Explicit 25-V window medians ranged from 16.5753468783 to 6,637.68690849 and 90th percentiles from 37.6841391646 to 16,766.9073404. Aggregated unregularized errors increased approximately linearly with noise (Table 6).

Table 6: Explicit daily $L = 5$ broadband injections, unregularized SVD. Each row pools $560 \times 32 = 17,920$ relative global errors.

Noise (V)	Median relative error	P90 relative error
10	58.9882	438.982
25	148.408	1,087.708
50	291.847	2,182.739
100	595.786	4,321.062

At 25 V, truncation at 10^{-3} reduced the pooled median and 90th percentile to 3.43809 and 6.66880, while ridge penalties 10^{-3} and 1 produced 0.846384/1.14231 and 0.864440/0.940035. These remain outside the 0.25/0.5 gate and are biased regularized estimates. The 10^{-9} truncated solution equaled the unregularized 148.408/1,087.708 result. Degree-shell challenges had unregularized medians from 146.842 to 148.708; the weakest-mode challenge gave 146.898 median and 1,061.56 at the 90th percentile. The failure is therefore not an accident of the broadband truth spectrum.

The localized caps additionally show that small sampled residuals can coexist with large global errors (Table 7). For the 15-degree equatorial cap, the median sampled residual was only

$$4.769007918593412 \times 10^{-6}\ \text{V},$$

while the median global error was 100 V. Because caps lie outside the fitted basis, these values are descriptive only.

7.5 Blocked recovery

Every shared-parameter training operator remained full rank: 36/36 for the static $L = 5$ class and 108/108 for the constant-plus-synodic $L = 5$ class. Table 8 summarizes the exact ranges of

Table 7: Out-of-basis cap diagnostics over 560 daily windows. Errors are V.

Center (ϕ, λ)	FWHM	Amplitude	Approx. bias	Est. median	Est. P90	Total median	Total P90
$(0, 0)$	15	1800.804	80.0530	59.9292	85124.9	100.000	85125.0
$(0, 0)$	30	903.185	41.0706	802.183	53429.1	803.234	53429.1
$(60, 0)$	15	1800.804	80.0530	8596.52	172007	8596.89	172007
$(60, 0)$	30	903.185	41.0706	14795.1	110723	14795.1	110723
$(-60, 180)$	15	1800.804	80.0530	7775.67	152644	7776.09	152644
$(-60, 180)$	30	903.185	41.0706	14675.4	107732	14675.5	107732

the 32-draw, 25-V matched-truth tests. Errors are small because hundreds of thousands of sweeps constrain a genuinely shared synthetic coefficient vector. Longitude holdouts are the hardest, particularly for the synodic class, but remain below 1 V sampled RMS.

Table 8: Shared-parameter blocked recovery. Ranges span all splits of the named type; global and test quantities are median/P90 error in V.

Model	Split	Condition	Global median	Global P90	Test median / P90
Static	mission tail	2.28099	0.297567	0.354559	0.279850 / 0.319962
Static	12 longitude sectors	4.12540–4.17478	0.260792–0.281630	0.309194–0.341257	0.300132–0.368343 / 0.453577–0.542919
Static	four seven-day folds	2.23083–2.30305	0.286837–0.312589	0.331306–0.347164	0.264512–0.289454 / 0.315540–0.319660
Synodic	mission tail	2.57313	0.516044	0.566709	0.483320 / 0.528202
Synodic	12 longitude sectors	4.28472–4.39554	0.478147–0.499358	0.521463–0.544529	0.587156–0.680886 / 0.732551–0.823409
Synodic	four seven-day folds	2.53311–2.64617	0.507485–0.530116	0.541629–0.560522	0.476686–0.495051 / 0.510225–0.527096

Daily blocked fits remained algebraically rank 36, yet global recovery was catastrophic. The finite median-global-error ranges were

$$\begin{aligned} \text{560 contiguous-time tests:} &\quad [2.78127999422 \times 10^3,\ 3.68024426671 \times 10^6]\ \text{V}, \\ \text{6,524 longitude-sector tests:} &\quad [1.23274522598 \times 10^3,\ 2.03174795607 \times 10^6]\ \text{V}. \end{aligned}$$

One contiguous-time covariance result was infinite. Low in-sample or held- footprint errors therefore do not imply recovery away from the thin daily tracks.

7.6 Sensitivity analyses

All 16 frozen daily $L = 5$ sensitivity records had zero stable windows, zero recovery-passing windows, and witness fraction one. Table 9 shows the principal native contrasts. The six paired common-sweep cohorts gave the same verdict, with median visibility between $1.9548336121415843 \times 10^{-5}$ and $4.2555408221077033 \times 10^{-5}$.

Representing all row footprints with total sweep weight one also failed, with median visibility $5.290155297417636 \times 10^{-5}$. The invalid false- precision analysis that treated row footprints as independent failed as well, with $4.632934509032125 \times 10^{-5}$. Thus the conclusion is insensitive to the reducer, complete-sweep selection, attitude lookup, production frame mix, common-cohort pairing, body-fixed versus solar co-rotating longitude, monopole removal, and row-representation convention. The coordinate-frame bug remains important for physical footprint provenance even though it does not rescue the daily inverse problem.

The retained empirical unit-column-normalized spectrum reduced the median daily condition number from 47,125.2 to 29,842.8 (maximum 1,010,653), but the physical visibility of its selected weakest mode remained 4.64097×10^{-5} median and 5.65089×10^{-4} maximum. It therefore changes the apparent coefficient collinearity without rescuing the physical field.

Table 9: Primary sensitivity summaries. Each contains 560 daily windows and fails the class gate.

Variant	Median condition	Median minimum visibility
Canonical mean	47,125.2	4.51915×10^{-5}
Canonical midpoint row	41,887.6	5.15730×10^{-5}
Complete 15-row geometry	99,715.2	2.18497×10^{-5}
Legacy next-sample attitude	49,285.0	4.25414×10^{-5}
Production frame-mixed	68,788.8	3.00415×10^{-5}
Submitted combined legacy	67,656.1	3.07093×10^{-5}
Solar co-rotating longitude	14,364.2	1.42608×10^{-4}
Contrast only (35 parameters)	43,906.3	4.79262×10^{-5}

8 Outcome and interpretation

For the primary daily $L = 5$ model, the unstable fraction, failed-recovery fraction, and material-witness fraction are all exactly 1.0. It therefore satisfies every frozen Outcome B criterion. However, multiple prospectively specified classes meet the at-least-90% stability and recovery requirements without a material witness: 72-hour $L = 1$; 168-hour $L \leq 3$; one-synodic- period $L \leq 10$; whole-mission static $L \leq 20$; and whole-mission shared constant-plus-synodic $L \leq 10$. Under the frozen precedence rule, the final classification is therefore **Outcome C**, not B.

Outcome C is a boundary statement. The daily $L = 5$ model is algebraically full rank but physically non-identifiable at the frozen 100-V field and 25-V sweep-noise scales. Longer independent windows and tightly shared temporal bases can become conditionally identifiable because they pool orbital coverage over time. Such pooling changes the estimand: it does not reconstruct an instantaneous global map.

9 Limitations

First, the experiment never reads a measured potential. Its iid Gaussian noise levels are controlled reference scales, not an empirical error model for loss-cone fits. Real errors may be heteroscedastic, correlated, biased, and selected by environment or fit quality. The oracle inclusion of every connected sweep makes failures strong but successes conditional.

Second, the spacecraft-potential gauge is unresolved. Absolute $\ell = 0$ surface potential and corresponding temporal terms cannot be separated from an unknown additive spacecraft potential using this operator alone. The contrast checks remove the exact gauge direction but do not supply a physical spacecraft-potential calibration.

Third, all passing models assume their declared band limit and temporal class. The localized-cap tests show severe aliasing when truth lies outside the daily $L = 5$ basis. Passing matched-truth blocked tests for a static or shared synodic model do not establish that the Moon obeys either model.

Fourth, canonical results correct a material production frame mismatch. Any previous product using the mixed-frame path requires separate reassessment of footprints, connectivity, polarity, and illumination. The robustness of the identifiability verdict across frame variants does not validate those physical locations.

Finally, the pass/fail thresholds are prospective engineering definitions of usable visibility and noisy recovery. The phrase "physical non- identifiability" here means failure under those declared

metrics; it does not erase the mathematical fact that every primary matrix has nonzero singular values in finite precision.

10 Conclusions

Actual Lunar Prospector geometry does not support the submitted independent daily degree-5 global field. Full numerical rank alone is misleading: a 100-V global weakest mode produces only 1.085×10^{-4} to 0.0405 V RMS on a day's sampled footprints, and 25-V noise drives global recovery errors many times the truth amplitude. This failure survives every frozen geometry, gauge, reducer, longitude, and row-weighting sensitivity.

The same geometry does support prospectively defined coarse independent windows, mission-integrated static fields, and a mission-shared constant-plus-synodic basis. Those successes establish conditional design identifiability and motivate sharply constrained summaries, not an instantaneous global potential map. Until the scalar measurement itself and the spacecraft-potential contribution are independently validated, Experiment 008 supports no claim of recovered physical lunar surface potential.